Prinson Royal Nazareth

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Summary

Master of Computer Applications (MCA) candidate specializing in **Python**, **TypeScript**, **full-stack web development**, and **autonomous AI agents**. Built and published open-source developer tools, CLI utilities, and backend systems with **1,450+ combined npm downloads**, emphasizing modular architecture, security sandboxing, and scalable software engineering.

Technical Skills

Languages: Python, TypeScript, JavaScript (ES6+), Java, C, SQL, HTML5, CSS3

Frameworks & Libraries: React.js, Node.js, Express.js, FastAPI, Typer, Tailwind CSS, GSAP

AI & Automation: Autonomous AI Agents, LLM Tool Use, Function Calling, AST Code Indexing, OS Automation

Databases & APIs: MongoDB, RESTful APIs, WebSockets, Server-Sent Events (SSE)

Developer Tools & Testing: Git, GitHub, GitHub Actions, Linux, Postman, pytest, npm

Core Competencies: Data Structures & Algorithms, Backend Development, Full-Stack Web Development, Systems Architecture

Projects

prinson.is-a.dev/projects

CODE68 | **Autonomous Terminal AI Coding Agent** *Python, FastAPI, Typer, AST, SSE, OpenAI API* | GitHub | Live Demo 2026

- Architected an **autonomous CLI coding agent** decoupling **remote GPU inference** (4-bit quantized Qwen2.5-Coder via FastAPI and SSE streaming) from **local execution**, eliminating local VRAM overhead across project files and Git workflows.
- Engineered a **multi-step tool-calling agent loop** featuring loop detection, **Python AST symbol extraction**, ripgrep search, and **surgical unified-diff code editing** across 4+ LLM backends (Kaggle, Ollama, Gemini, OpenRouter).
- Implemented a **5-tier security sandbox** with path-traversal guards, destructive command interception, secret redaction, and audit logging, verified by a **27-test automated pytest suite**.

PRIVACY68 | **Privacy-First AI Desktop Automation Agent** *Python, Function Calling, AI Agents, Local Tool Execution* Sep. 2026 – Present

- Built a **local-first Python AI agent** using **structured function calling** to automate OS-level workflows and desktop tasks with zero external user-data exposure.
- Integrated **real-time web search** with **sandboxed local tool execution** to accelerate information retrieval and system automation.
- Designed a **pluggable modular architecture** enabling rapid addition of custom AI tools while isolating core execution logic.

SWARM66 | **Autonomous DOM Cursor Swarm Engine** *TypeScript, JavaScript, HTML5, Reynolds Boids Physics* | GitHub | Live Demo 2026

- Engineered a **zero-dependency DOM visualization engine** that reverse-engineers webpage layouts in real time, constructing UI elements from a blank canvas via **18 autonomous cursors** driven by **Reynolds Boids flocking physics**.
- Developed **4-agent cursor squads** with live pixel dimension badges, **5+ framework integrations** (React, Next.js SSR, Vue, Angular, Svelte), and a **360-degree orbit convergence** animation, reaching **1,027+ peak weekly npm downloads** (**@leozodiac/swarm66**).

WEB66 | **Codebase Intelligence & Dependency Visualizer** *Node.js, JavaScript, WebSockets, Canvas 2D* | GitHub 2026

- Created a **zero-dependency Node.js CLI** that recursively parses TypeScript, JavaScript, and Python codebases into a **node/edge dependency graph**, paired with a **WebSocket server** for debounced file watching and **live browser hot-reload**.
- Built an **interactive Canvas 2D graph visualizer** with force-directed physics and animated energy pulses, achieving **435+ peak npm downloads** as **@leozodiac/web66** (**npx web66**).

AJ-ASTRIX 2026 | **College Fest Web Application** *React.js, Vite, Tailwind CSS, GSAP* Apr. 2026

- Developed a **responsive multi-page web application** delivering real-time event schedules, rules, and registration workflows for college fest participants.
- Built **reusable React components** and hardware-accelerated **GSAP animations** styled with Tailwind CSS to maximize cross-device usability.

Personal Portfolio Website *React.js, Vite, Tailwind CSS* | Live Demo Feb. 2026

- Designed and deployed a **responsive developer portfolio** showcasing 6+ full-stack projects, open-source npm packages, and technical skills across desktop and mobile viewports.

Education

Master of Computer Applications (MCA) 2025 – 2027
AJ Institute of Engineering & Technology, Mangaluru *Currently Pursuing*

Bachelor of Computer Applications (BCA) 2023 – 2025
Samvedana Science and Commerce First Grade College

Certifications

- **Python for Data Science** | NPTEL (4-Week Program)
- **C Programming Basics** | Simplilearn SkillUp